

C ++ Programming Assessment Test

- Create a Project to demonstrate an ATM Banking Application which will help to get to know the bank information.. The Project should contain all the necessary requirements i.e A Program should use the concept of OOP.(compulsory : Constructor),Make use of scope resolution operator, Loops, ConditionalStatements, Whichever is required.
- Give comments where it is required , for better understanding of codes. It will help you and the project evaluator.
- A Project should contain the following execution flow. 1. The very first task is to Display the Welcome SCREEN for the User. It should display the current Date and Time A Menu to display the choice, Consider ATM PIN as 12345

Answer:

```
#include <iostream>

#include <ctime>    // For date and time

using namespace std;

// ATM Class Definition

class ATM {

private:

    int pin;

    double balance;

public:

    // Constructor (Compulsory)

    ATM();

    void displayWelcome();

    void verifyPin();

    void showMenu();

    void checkBalance();
```

```

    void deposit();

    void withdraw();

};

// Constructor Definition using Scope Resolution Operator
ATM::ATM() {

    pin = 12345;    // Default PIN

    balance = 5000.0; // Initial Balance

}

// Function to Display Welcome Screen
void ATM::displayWelcome() {

    time_t now = time(0);

    char* dt = ctime(&now);

    cout << "===== " << endl;

    cout << "    WELCOME TO ATM SYSTEM    " << endl;

    cout << "===== " << endl;

    cout << "Current Date & Time: " << dt;

    cout << "===== " << endl;

}

// Function to Verify PIN
void ATM::verifyPin() {

    int enteredPin;

    int attempts = 0;

    while (attempts < 3) {

```

```

cout << "Enter your ATM PIN: ";
cin >> enteredPin;

if (enteredPin == pin) {
    cout << "PIN Verified Successfully!\n";
    showMenu();
    return;
} else {
    cout << "Incorrect PIN. Try Again.\n";
    attempts++;
}
}

cout << "Too many wrong attempts. Card Blocked!\n";
}

// Function to Show ATM Menu
void ATM::showMenu() {
    int choice;

    do {
        cout << "\n===== ATM MENU =====\n";
        cout << "1. Check Balance\n";
        cout << "2. Deposit Money\n";
        cout << "3. Withdraw Money\n";
        cout << "4. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;
    } while (choice < 1 || choice > 4);
}

```

```

switch (choice) {
case 1:
    checkBalance();
    break;
case 2:
    deposit();
    break;
case 3:
    withdraw();
    break;
case 4:
    cout << "Thank you for using ATM. Goodbye!\n";
    break;
default:
    cout << "Invalid Choice! Please try again.\n";
}

} while (choice != 4);
}

// Function to Check Balance
void ATM::checkBalance() {
    cout << "Your Current Balance is: Rs. " << balance << endl;
}

// Function to Deposit Money
void ATM::deposit() {

```

```
double amount;

cout << "Enter amount to deposit: ";

cin >> amount;

if (amount > 0) {
    balance += amount;

    cout << "Amount Deposited Successfully!\n";

    cout << "Updated Balance: Rs. " << balance << endl;
} else {
    cout << "Invalid Amount!\n";
}
}
```

// Function to Withdraw Money

```
void ATM::withdraw() {
    double amount;

    cout << "Enter amount to withdraw: ";

    cin >> amount;

    if (amount > balance) {
        cout << "Insufficient Balance!\n";
    }

    else if (amount <= 0) {
        cout << "Invalid Amount!\n";
    }

    else {
        balance -= amount;

        cout << "Please collect your cash.\n";
    }
}
```

```
        cout << "Remaining Balance: Rs. " << balance << endl;
    }
}
```

```
// Main Function
```

```
int main() {
    ATM obj;          // Object Creation (Constructor Called)
    obj.displayWelcome(); // Welcome Screen
    obj.verifyPin();   // PIN Verification

    return 0;
}
```